Conducting system of heart

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Conducting System



Conducting System of the Heart (Cardiac Conduction System)

The **conducting system of the heart** is a specialized network of cardiac muscle cells responsible for generating and transmitting electrical impulses that coordinate atrial and ventricular contractions.

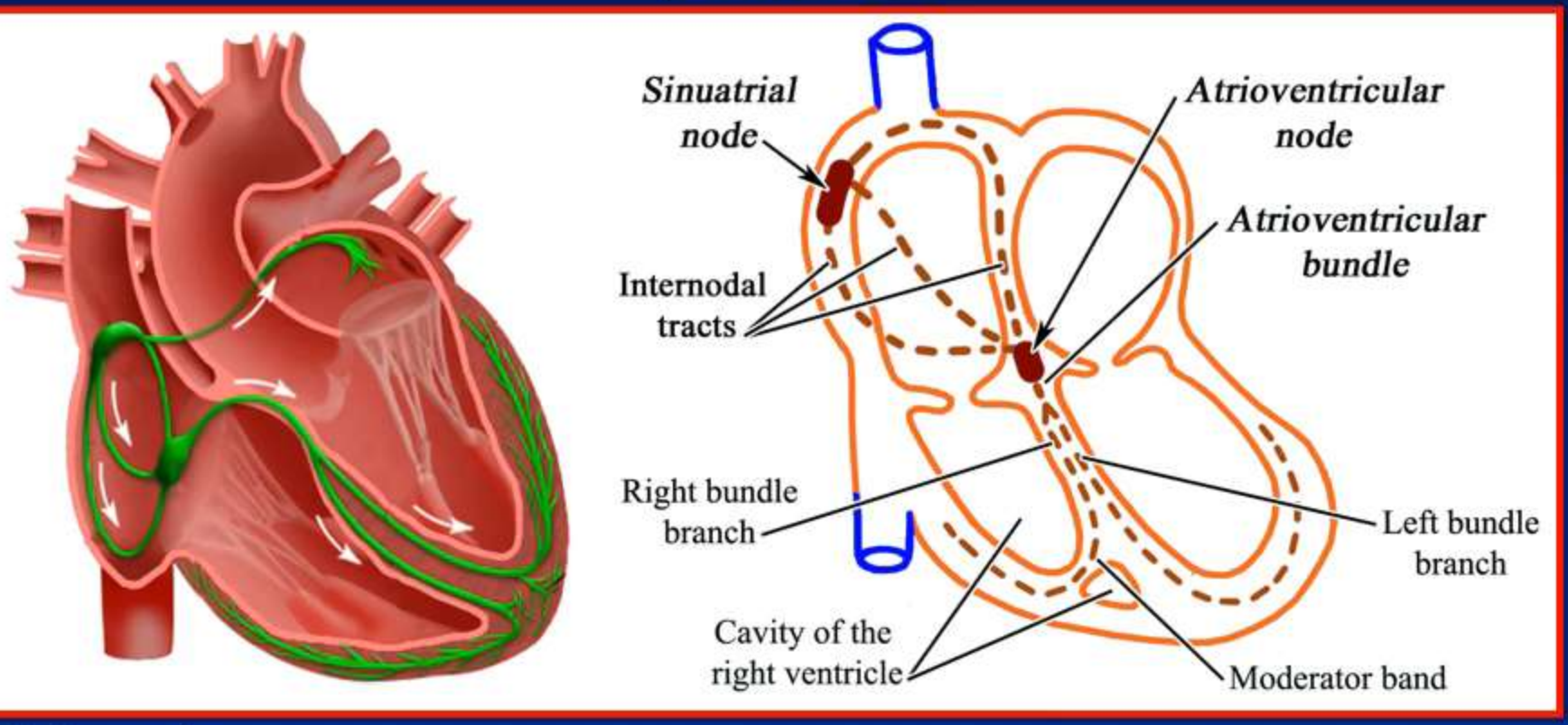
Its function is to ensure that:

.1 Both atria contract first.

.2 Both ventricles contract afterward.

.3 Ventricular contraction occurs in a coordinated manner to pump blood efficiently.

Parts



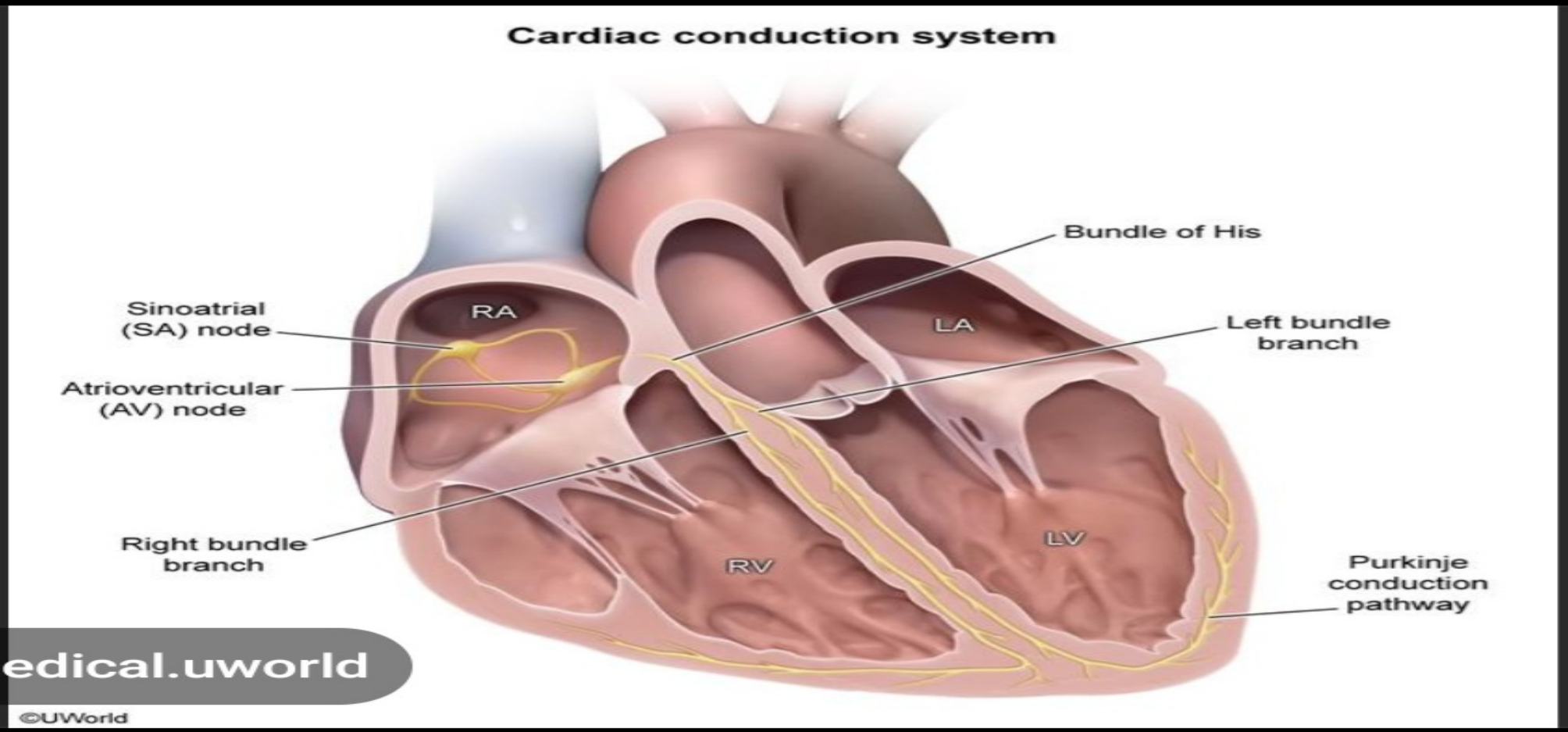
- S A node
- Internodal tractas
- A V node
- A V bundle

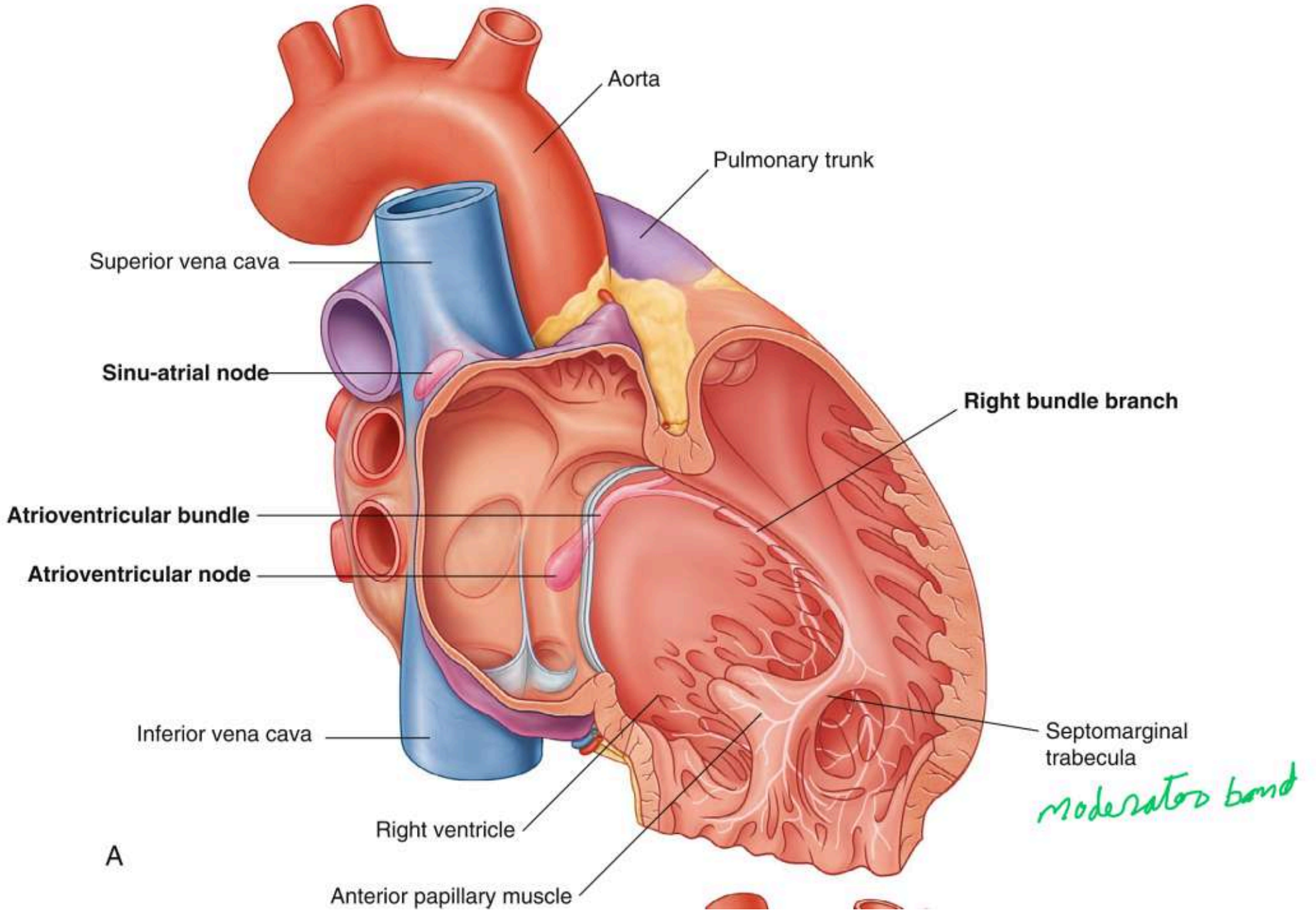
  a. Rt. bundle branch

  b. Lt. bundle branch

1. Components of the conducting system

There are five main components:




Aorta
Pulmonary trunk
Superior vena cava
Sinu-atrial node
Right bundle branch
Atrioventricular bundle
Atrioventricular node
Inferior vena cava
Septomarginal trabecula
moderator band
Right ventricle
A
Anterior papillary muscle

1. Sinoatrial node (SA node)

Natural pacemaker

- Location: Upper part of the right atrium, near the opening of the superior vena cava.
- Generates electrical impulses spontaneously.
- Normal firing rate: 60–100 impulses/minute.
- Initiates each normal heartbeat.

2. Atrioventricular node (AV node)

- Location: Lower part of the interatrial septum, near the opening of the coronary sinus and septal leaflet of the tricuspid valve.
- Receives impulses from the SA node.
- Delays conduction approximately 0.1 second.

Importance of AV nodal delay:
Allows the atria to empty blood into the ventricles before ventricular contraction.

Intrinsic rate: 40–60 beats/minute.

3. Bundle of His (AV bundle)

- Arises from the AV node.
- Passes through the fibrous skeleton of the heart.
- Enters the interventricular septum.
- Divides into right and left bundle branches.

4. Right and left bundle branches

- Run along the interventricular septum.
- Carry electrical impulses toward the apex of the heart.
- Left bundle branch divides into:
  - Left anterior fascicle.
  - Left posterior fascicle.

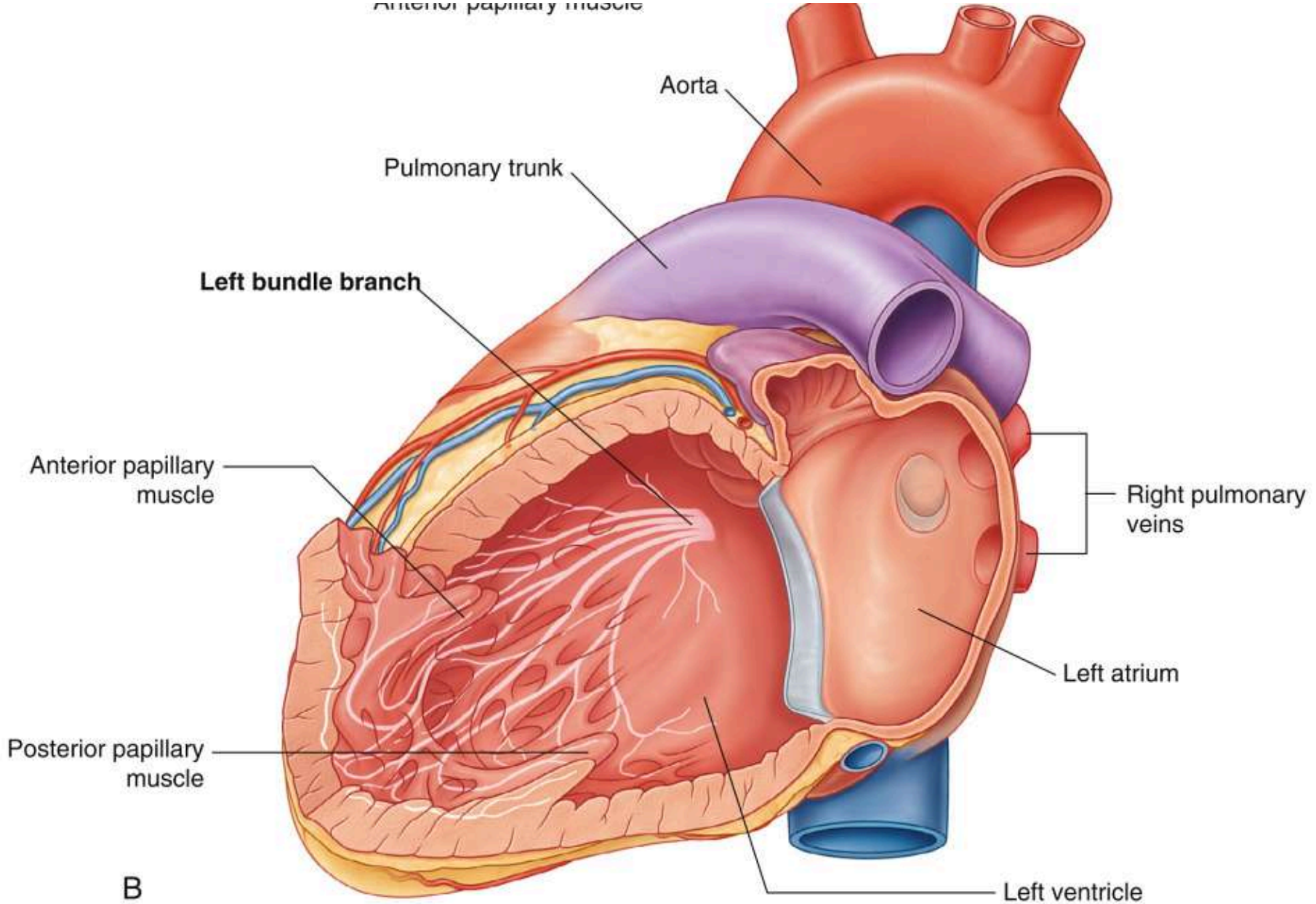
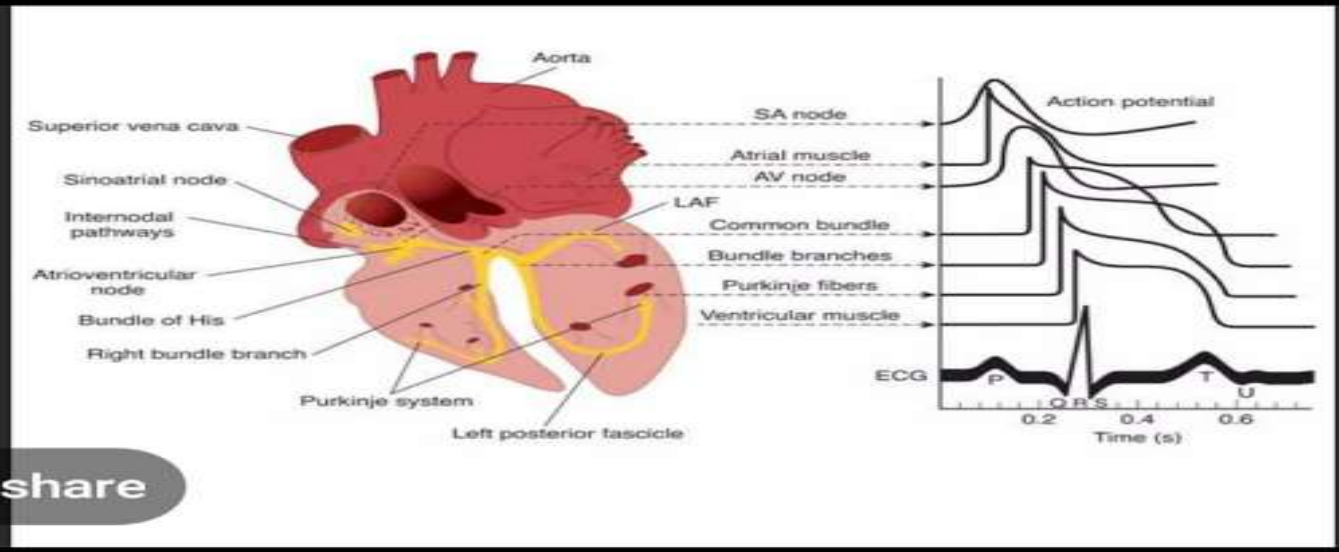


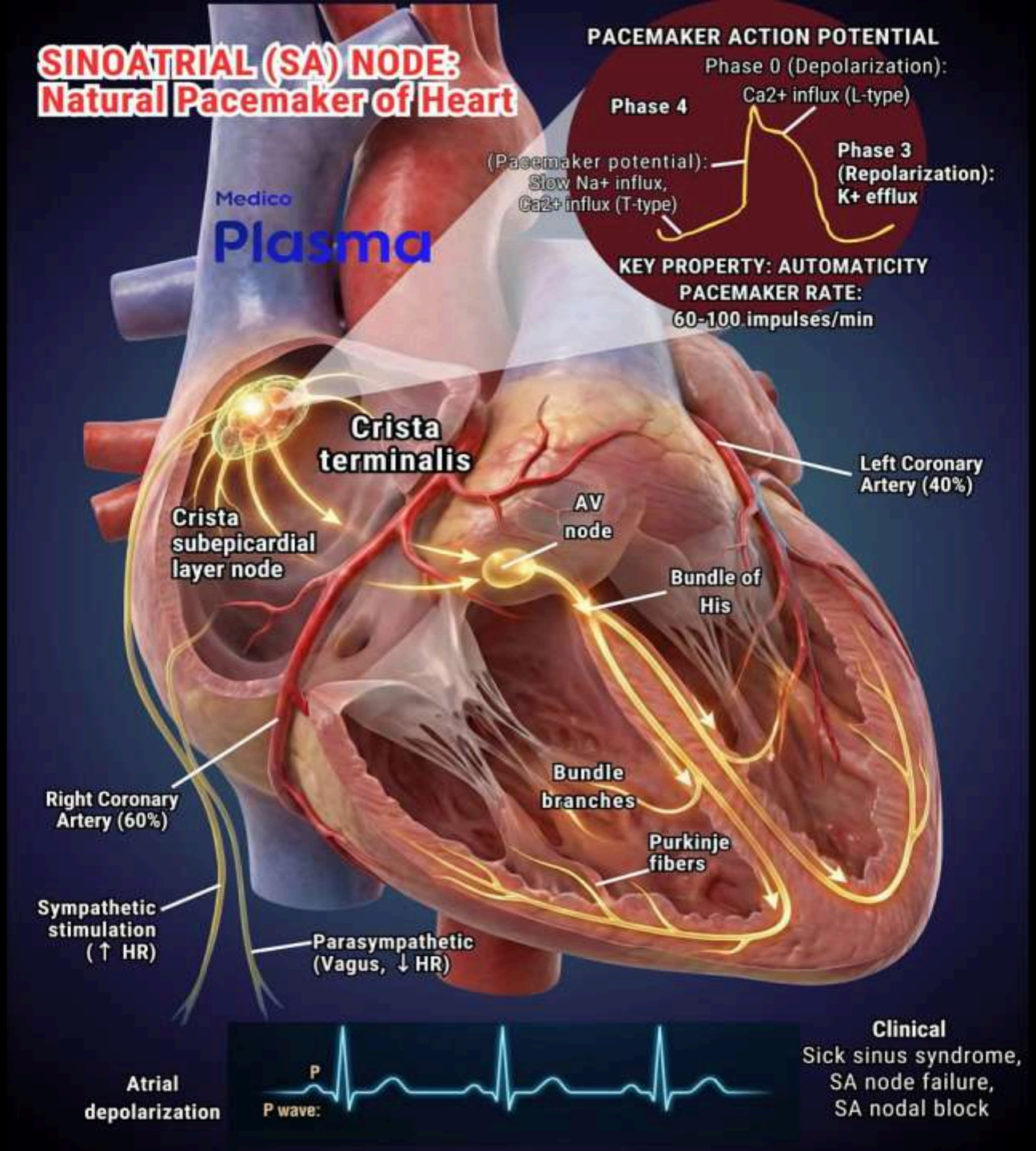
Fig. 3.86 Conduction System of the Heart. (**A**) Right chambers. (**B**) Left chambers.

5. Purkinje fibers

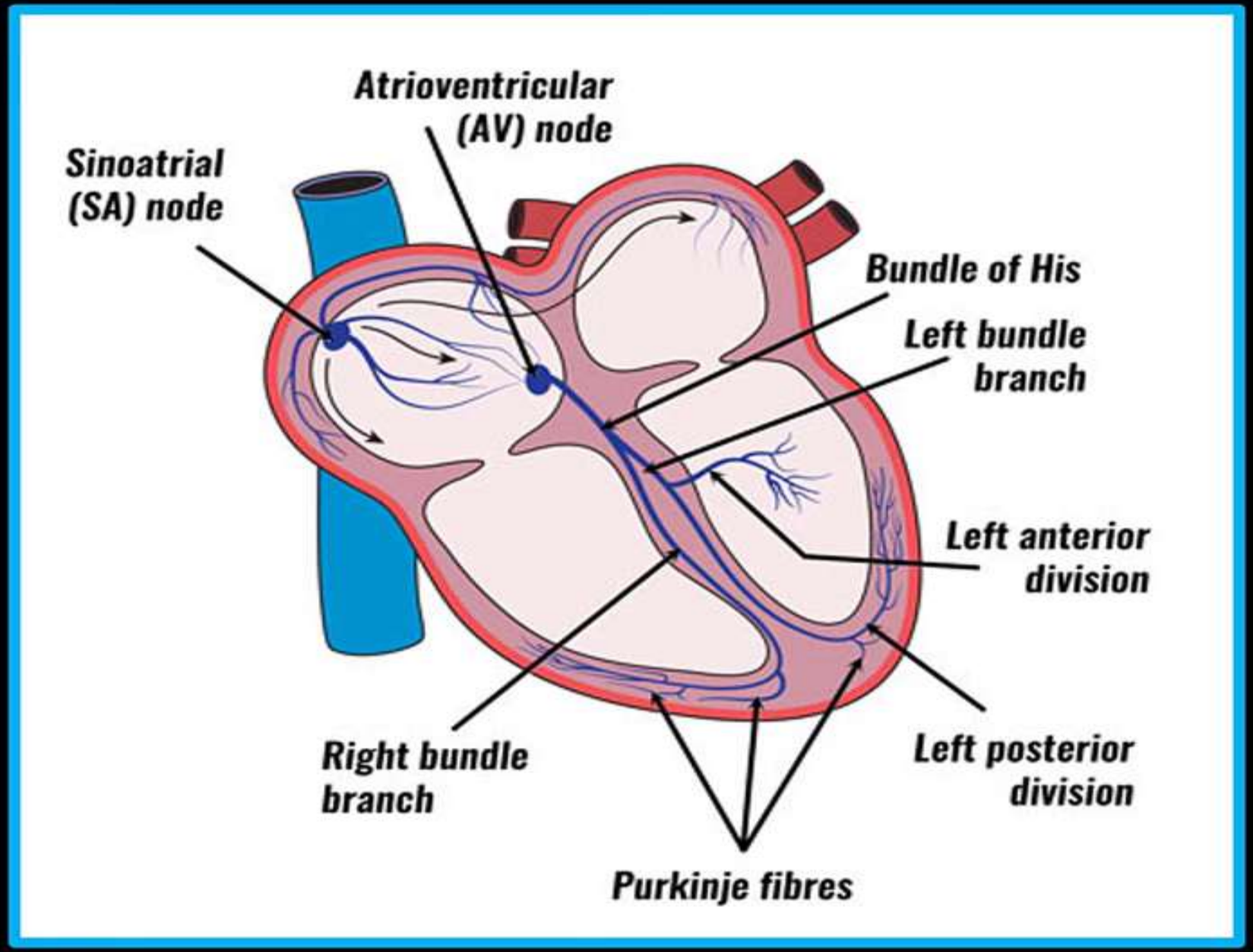


- Terminal branches of the conduction system.
- Distributed throughout the ventricular myocardium.
- Produce rapid electrical conduction.
- Cause coordinated ventricular contraction, generally from the apex upward.

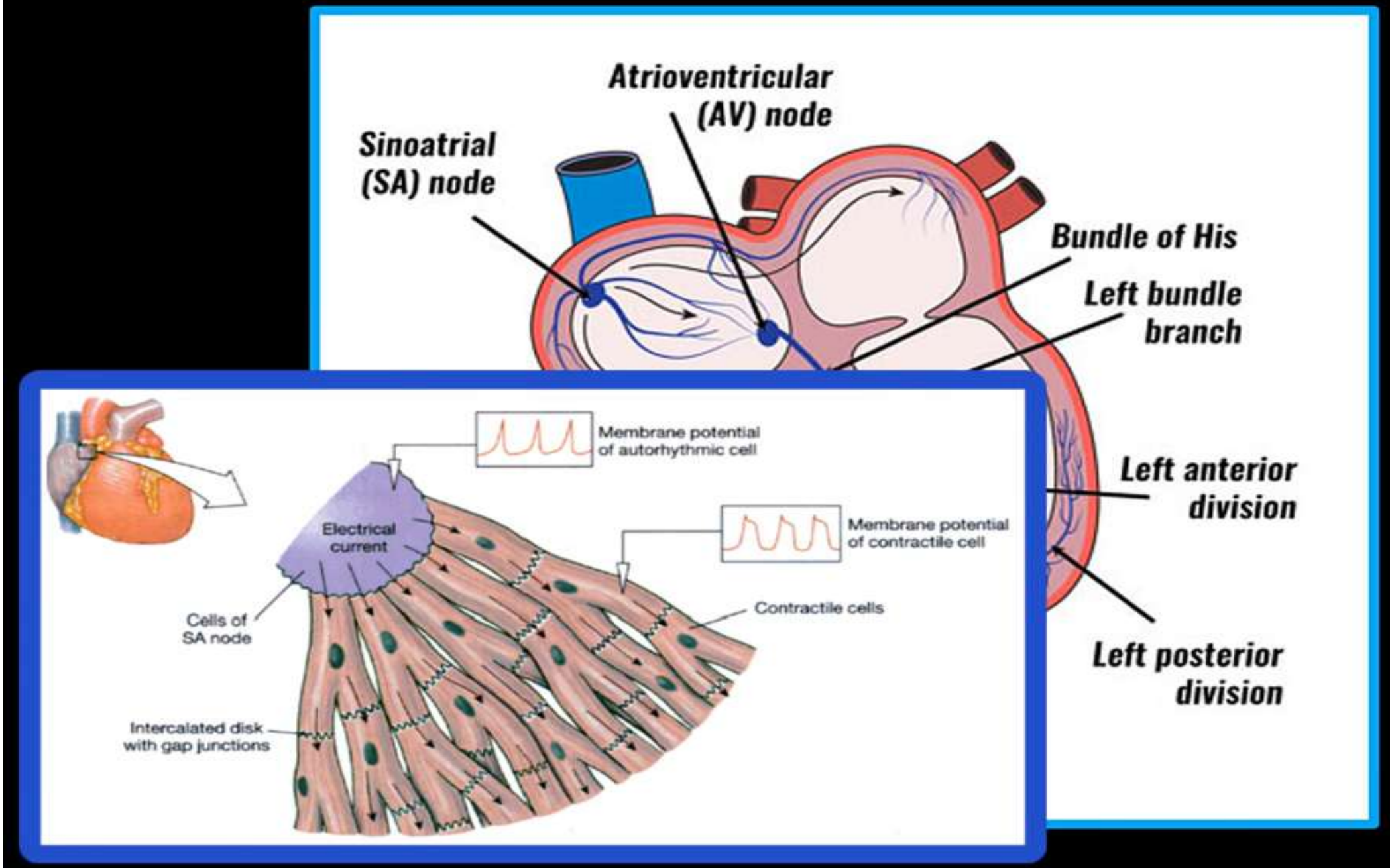
Intrinsic rate: 20–40 beats/minute.


SINOATRIAL (SA) NODE:
Natural Pacemaker of Heart
Medico
Plasma
PACEMAKER ACTION POTENTIAL
Phase 0 (Depolarization):
Ca2+ influx (L-type)
Phase 4
(Pacemaker potential):
Slow Na+ influx,
Ca2+ influx (T-type)
Phase 3
(Repolarization):
K+ efflux
KEY PROPERTY: AUTOMATICITY
PACEMAKER RATE:
60-100 impulses/min
Crista terminalis
Crista subepicardial layer node
AV node
Left Coronary Artery (40%)
Bundle of His
Bundle branches
Purkinje fibers
Right Coronary Artery (60%)
Sympathetic stimulation (↑ HR)
Parasympathetic (Vagus, ↓ HR)
Atrial depolarization
P
P wave:
Clinical
Sick sinus syndrome,
SA node failure,
SA nodal block

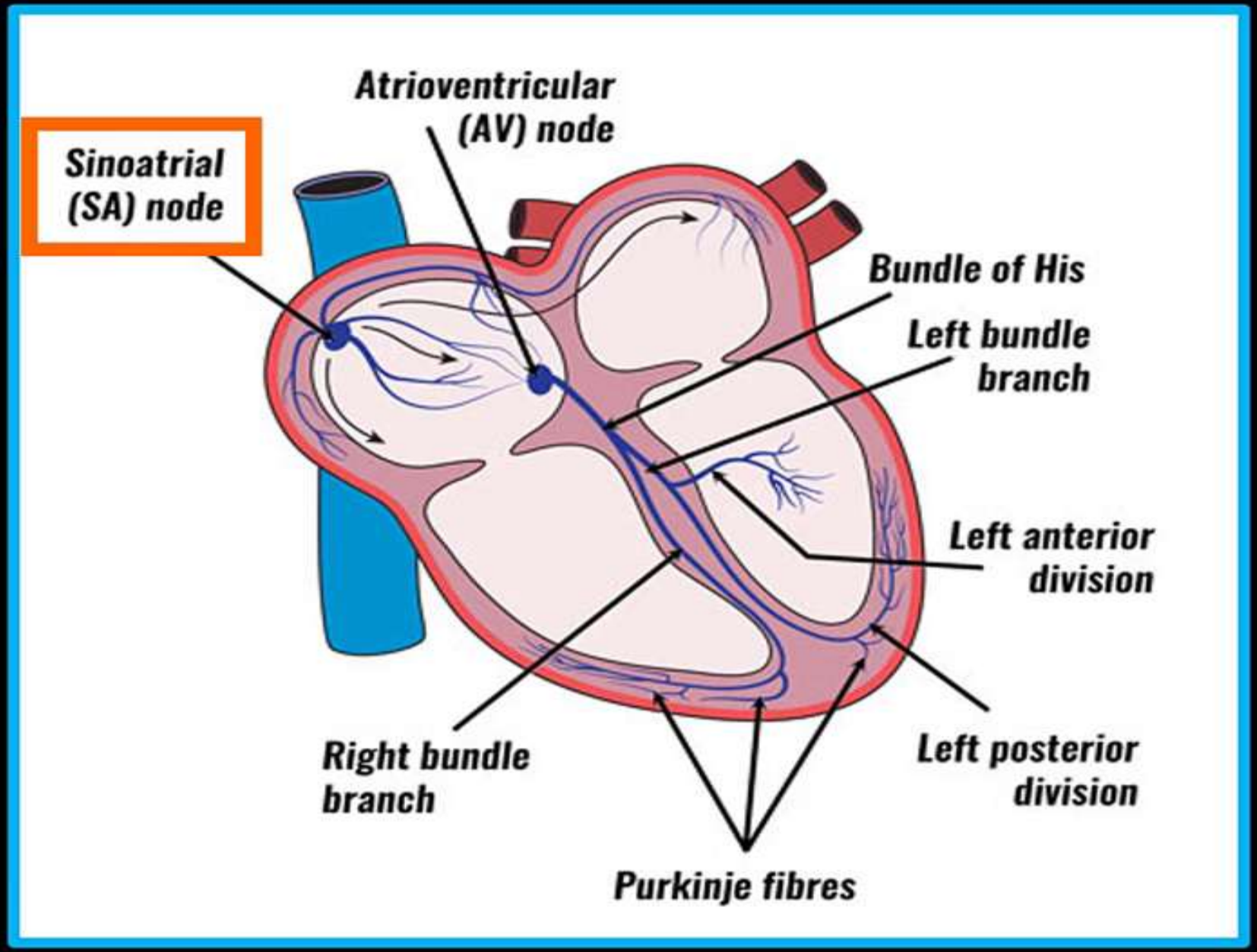
Cardiac Conduction System



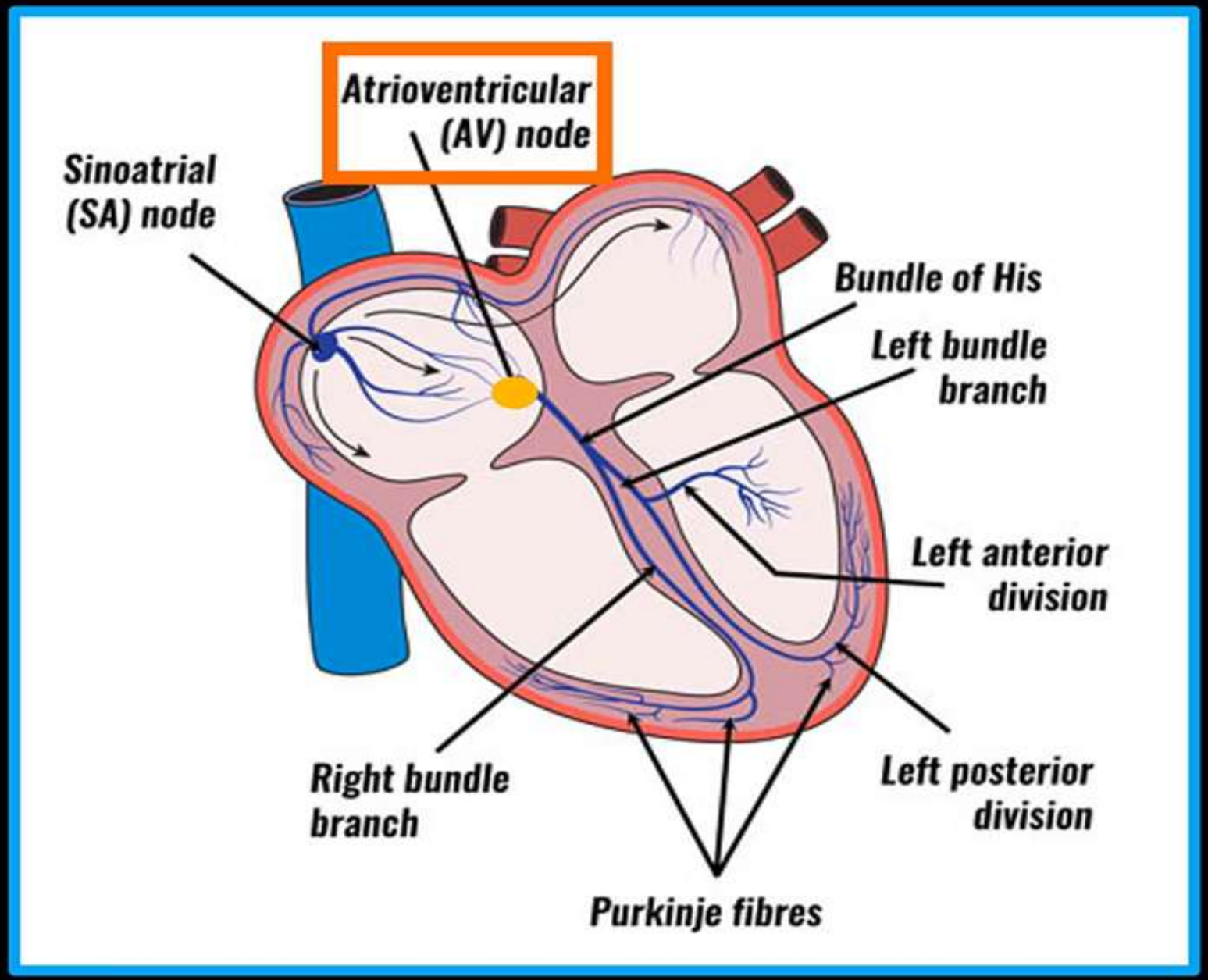
Cardiac Conduction System



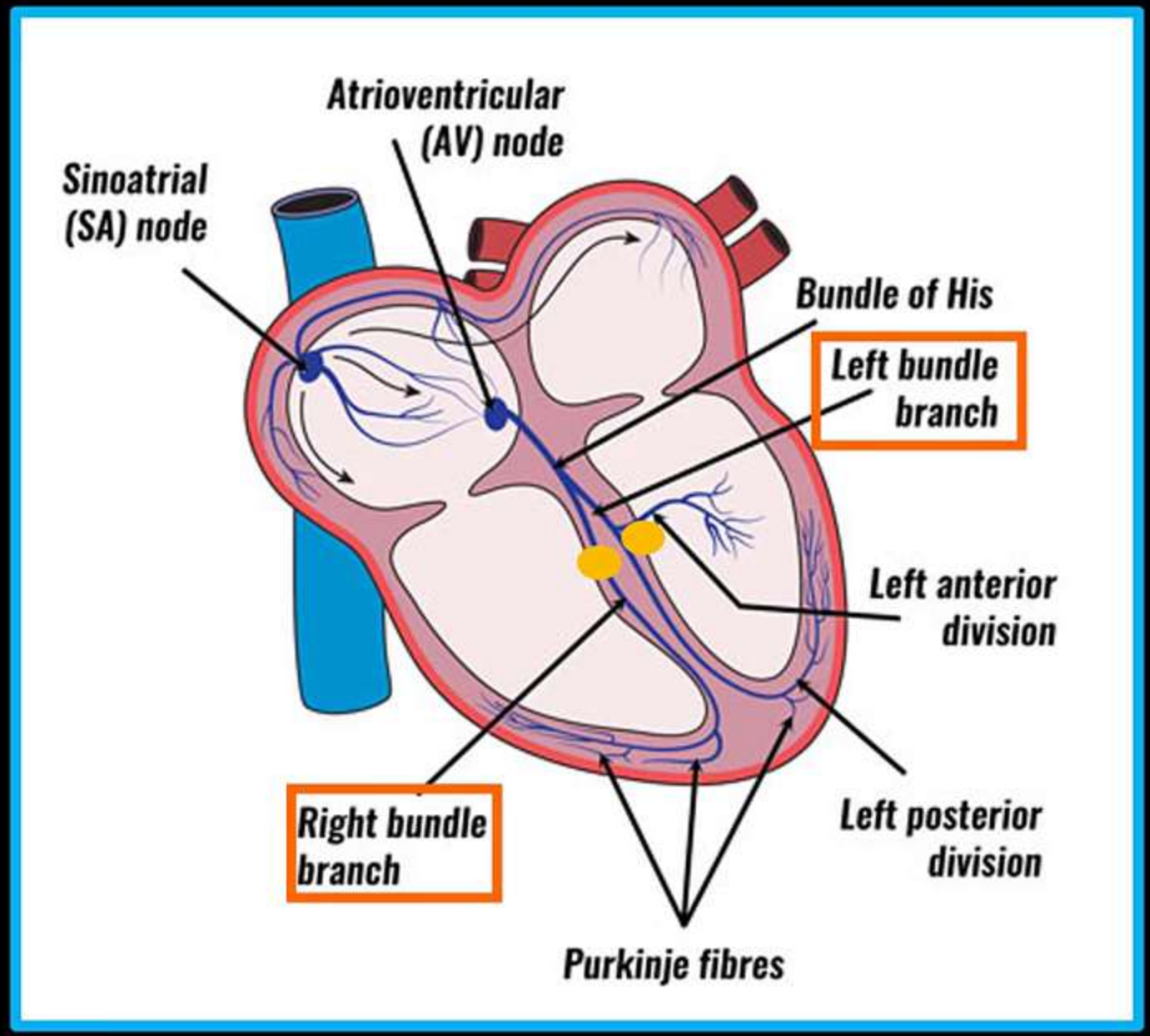
Cardiac Conduction System



Cardiac Conduction System



Cardiac Conduction System



Electrical conduction system of the heart:

1. Sinoatrial (SA) Node: Often called the heart's natural pacemaker, it initiates the electrical impulse.

2. Bachmann's Bundle (Interatrial Tract): Conducts the impulse from the right atrium to the left atrium.

3. Anterior Internodal Tract: One of three pathways connecting the SA node to the AV node.

4. Middle Internodal Tract (Wenckebach's pathway): The second internodal pathway.

5. Posterior Internodal Tract (Thorel's pathway): The third internodal pathway.

6. Atrioventricular (AV) Node: Acts as a gatekeeper, briefly delaying the signal to allow the ventricles to fill with blood.

7. Bundle of His (Atrioventricular Bundle): The bridge that carries the signal from the atria into the ventricles.

8. Right Bundle Branch: Carries the electrical impulse to the right ventricle.

9. Left Bundle Branch: Carries the electrical impulse to the left ventricle.

10. Left Anterior Fascicle: A branch of the left bundle that stimulates the upper/front part of the left ventricle.

10. Left Anterior Fascicle: A branch of the left bundle that stimulates the upper/front part of the left ventricle.

11. Left Posterior Fascicle: A branch of the left bundle that stimulates the lower/back part of the left ventricle.

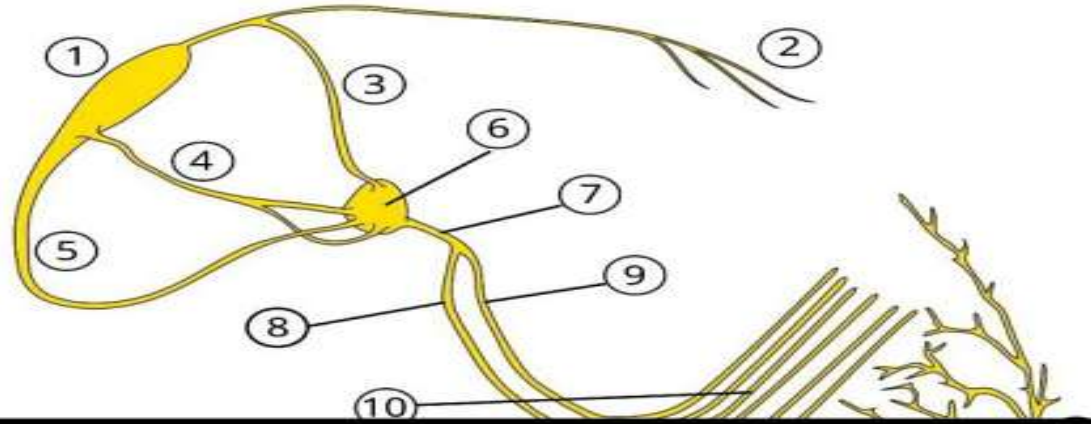
12. Purkinje Fibers: The final network of fibers that spread the impulse throughout the ventricular muscle, causing contraction.

》 Short debate questions:

"Are the classical anterior, middle, and posterior internodal tracts real structures, or just functional pathways in atrial myocardium?"

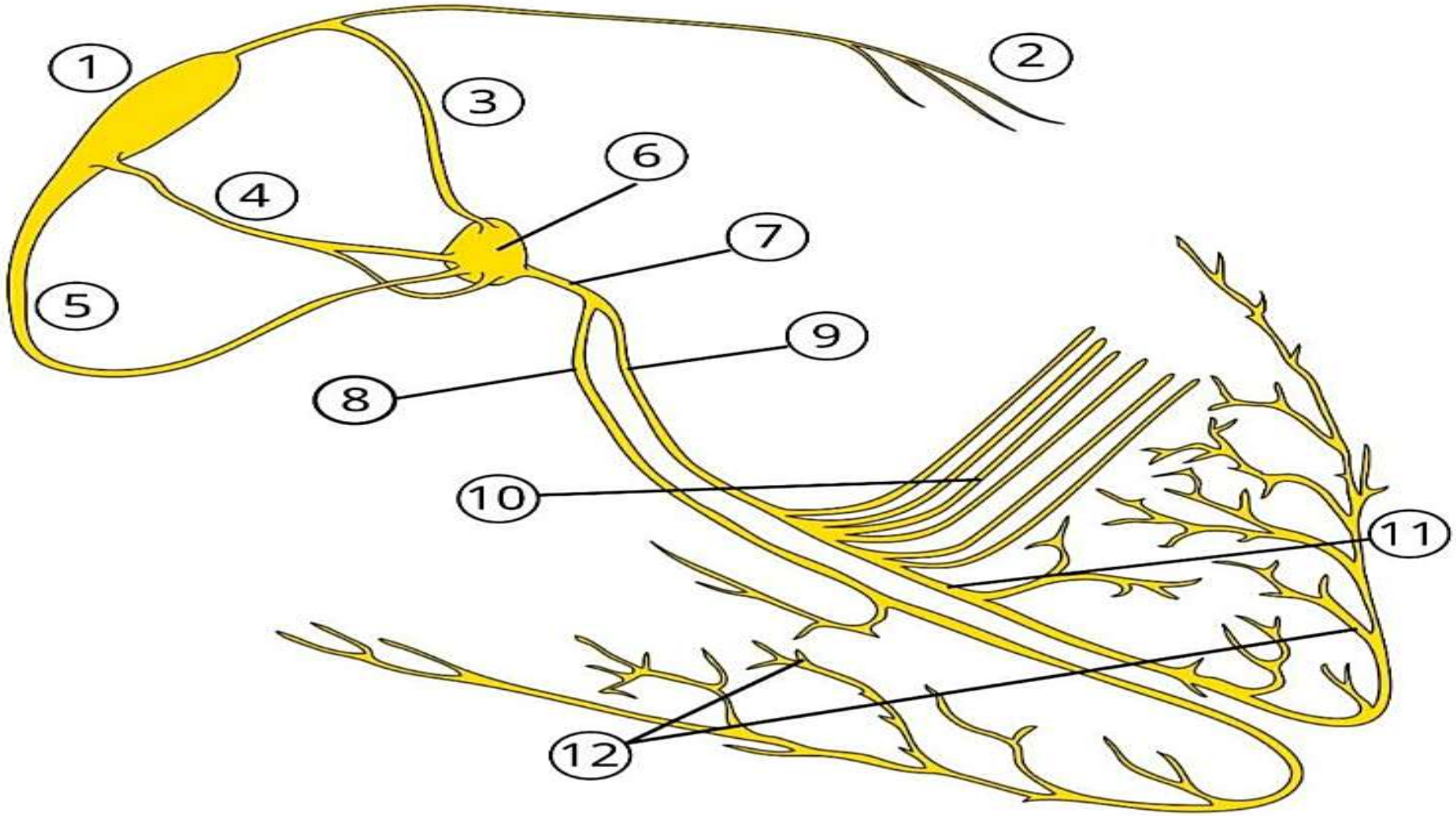
"If discrete internodal tracts are not clearly proven anatomically, how does the impulse travel so quickly from the Sinoatrial Node to the Atrioventricular Node?"

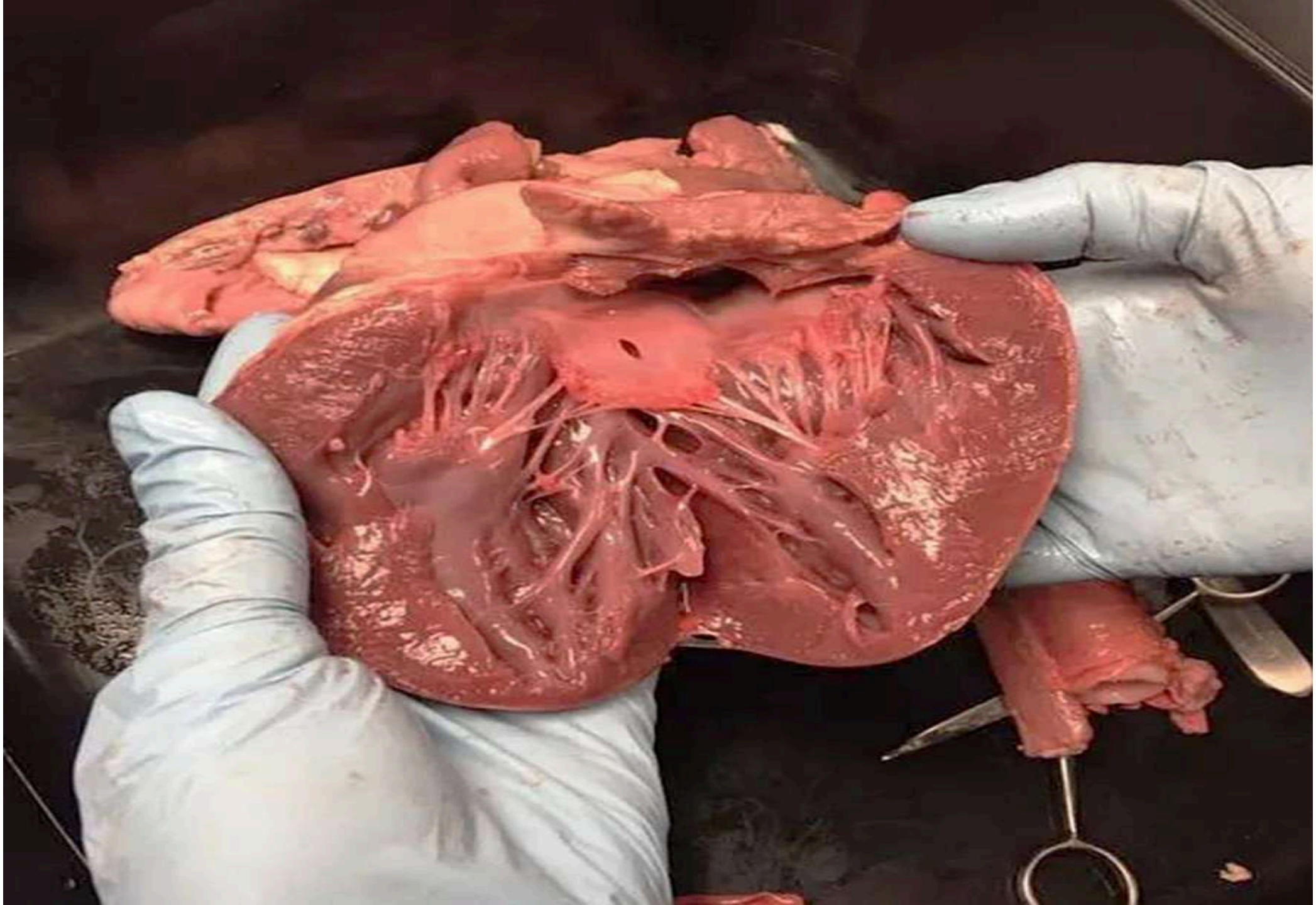
#lifestyle



anatomically, how does the impulse travel so quickly from the Sinoatrial Node to the Atrioventricular Node?”

#lifestyle





2. Sequence of electrical conduction

SA node
Impulse generation

↓

Atrial myocardium
Atrial contraction

↓

AV node
Electrical delay (~0.1 sec)

↓

Bundle of His

↓

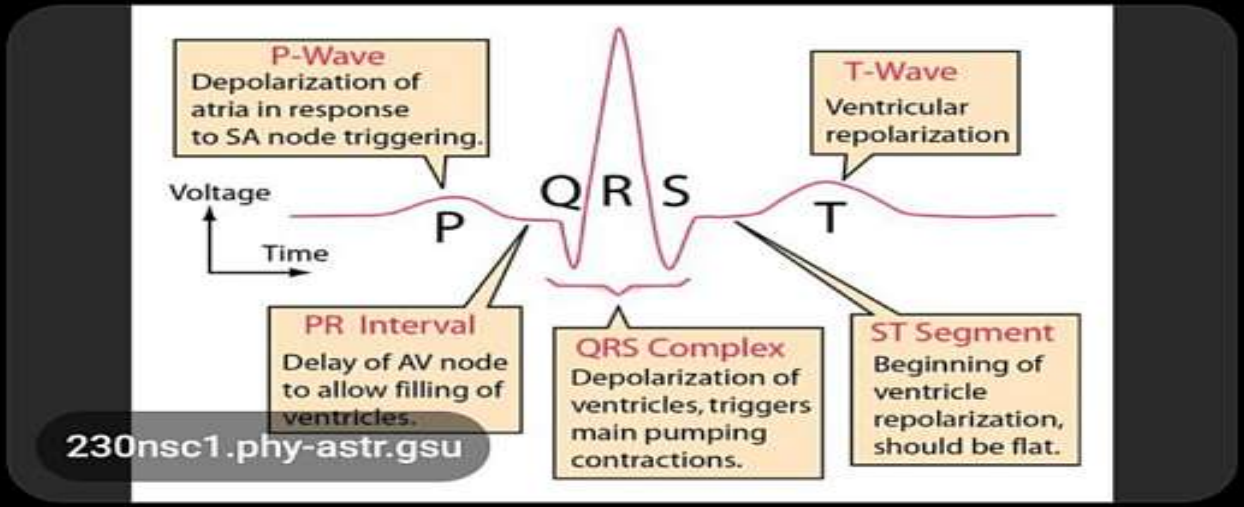
Right and left bundle branches

↓

Purkinje fibers
Ventricular contraction

3. Relationship with ECG

Very important for surgical and medical examinations.



ECG component	Electrical event
P wave	Atrial depolarization
PR interval	Conduction from atria through AV node to ventricles
QRS complex	Ventricular depolarization
T wave	Ventricular repolarization

Important normal values:

- PR interval: 0.12–0.20 seconds.
- QRS duration: Less than 0.12 seconds.
- Heart rate: 60–100 beats/minute.

4. Clinical importance

Abnormality	Explanation
Sinus bradycardia	SA node fires slowly
Sinus tachycardia	SA node fires rapidly
First-degree AV block	Prolonged PR interval
Second-degree AV block	Some atrial impulses fail to reach ventricles
Third-degree AV block	Complete dissociation between atria and ventricles
Right bundle branch block	Delayed right ventricular activation
Left bundle branch block	Delayed left ventricular activation

Cardiac innervation

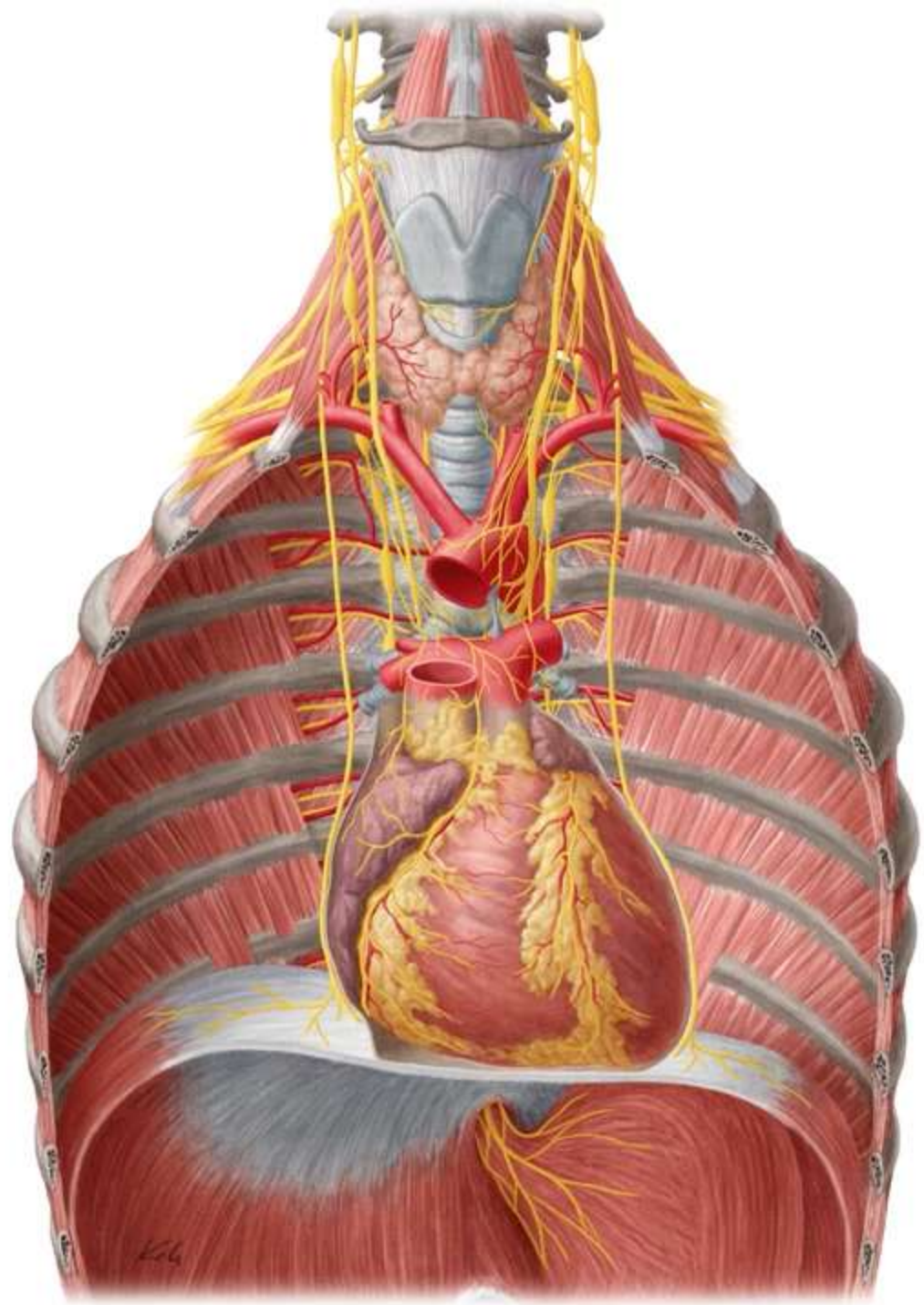
The autonomic division of the peripheral nervous system is directly responsible for regulating:

- heart rate,
- force of each contraction, and
- cardiac output.

Autonomic nervous system

Afferent - arrive

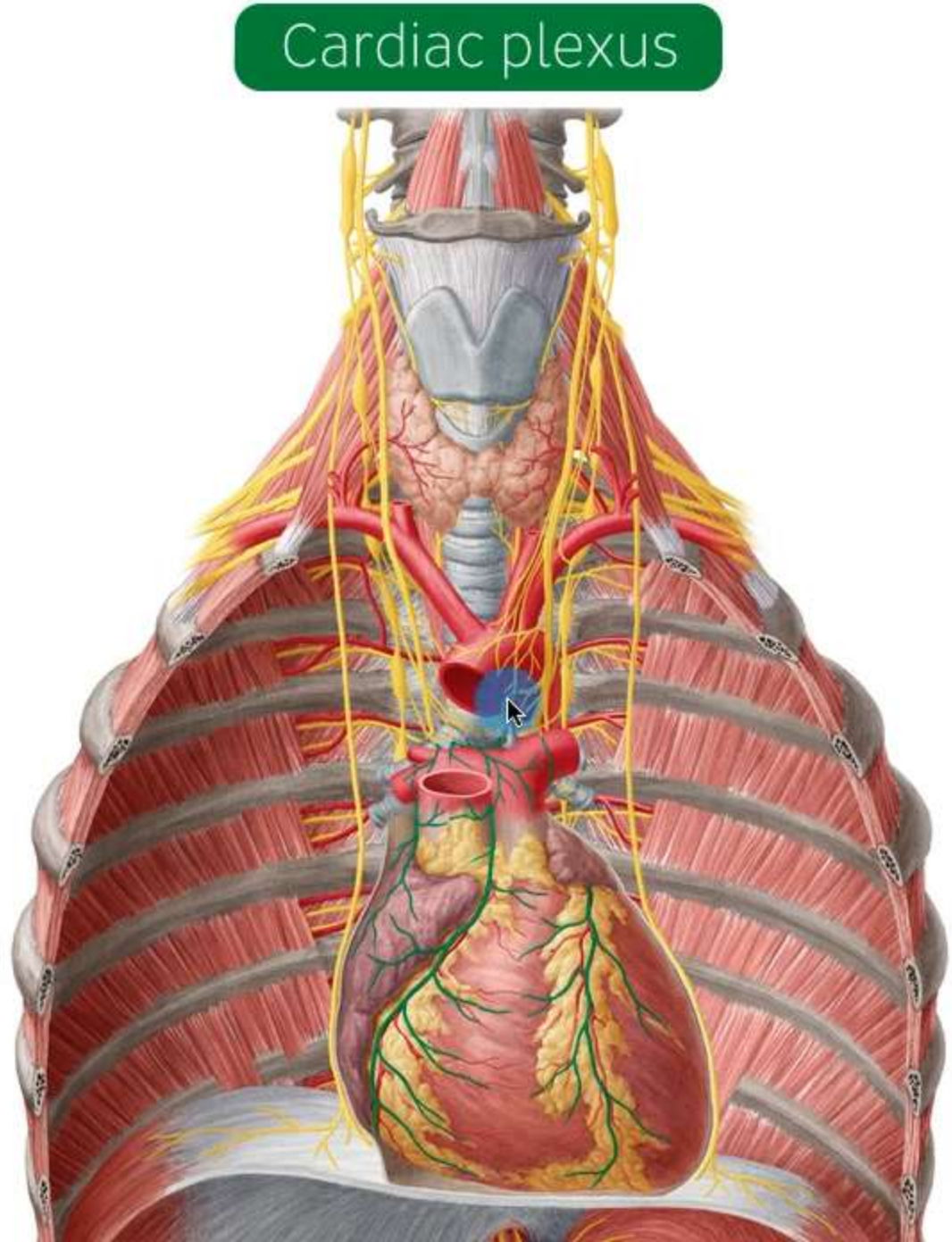
Efferent - exit



Autonomic fibers

Cardiac plexus is formed by cardiac nerves.

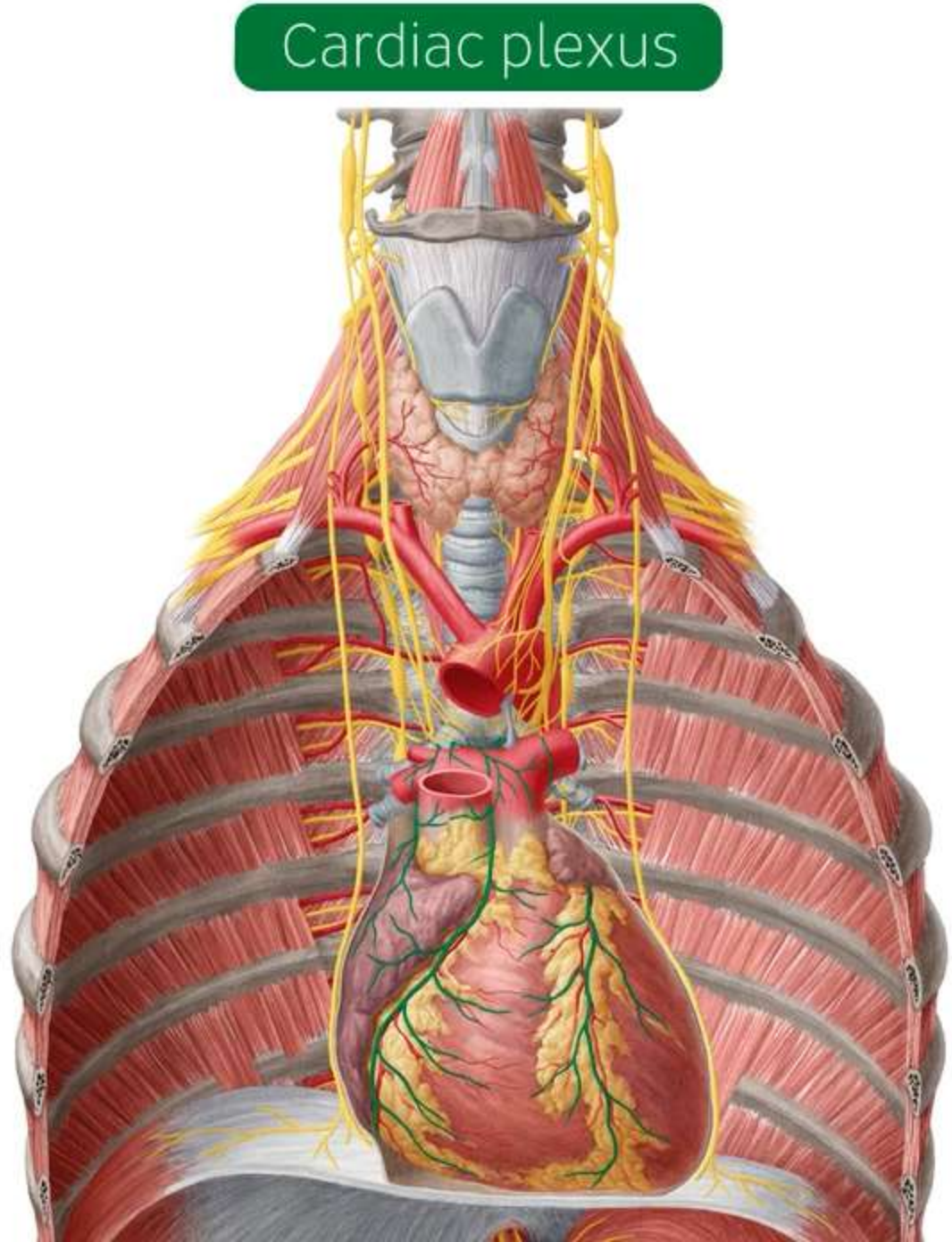
Sympathetic and parasympathetic fibres.



Function:

Innervate

- Electrical conducting system of the heart
- Atrial and ventricular myocardium
- Coronary vasculature



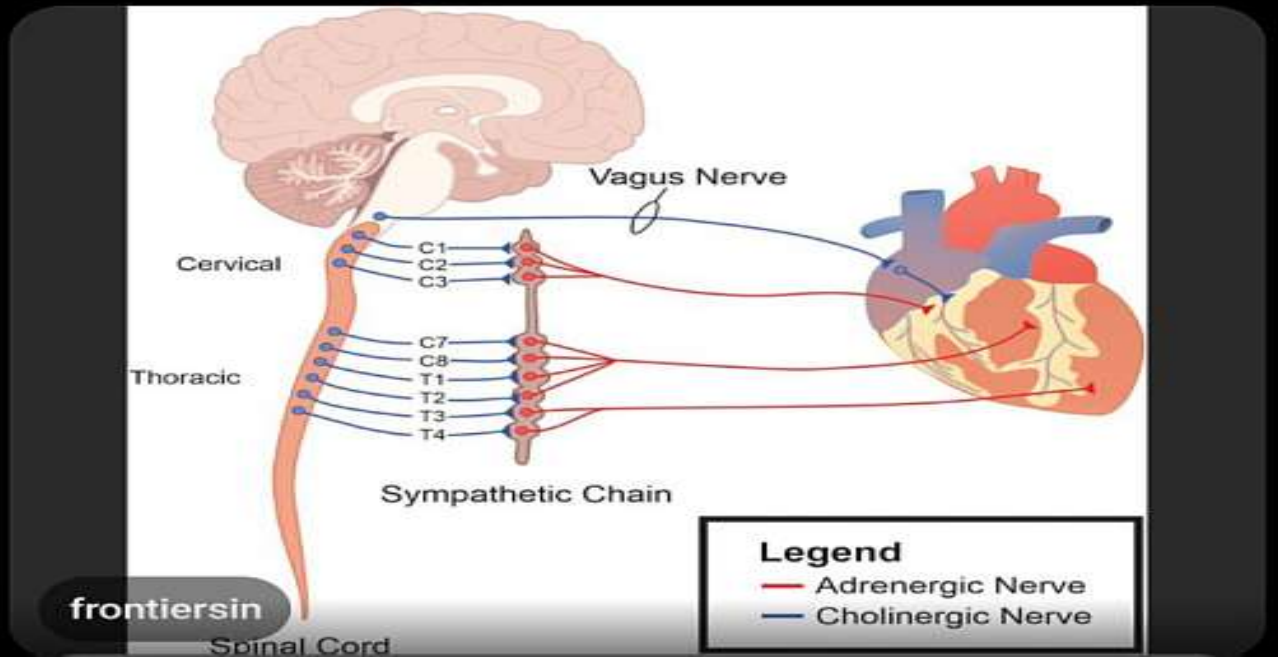
Innervation of the Heart (Cardiac Innervation)

The heart receives its nerve supply from the **Autonomic Nervous System (ANS)**, which consists of:

.1 Sympathetic nerves.
.2 Parasympathetic nerves (vagus nerve).

These nerves regulate heart rate, force of contraction, and conduction velocity.

1. Sympathetic innervation



Origin:

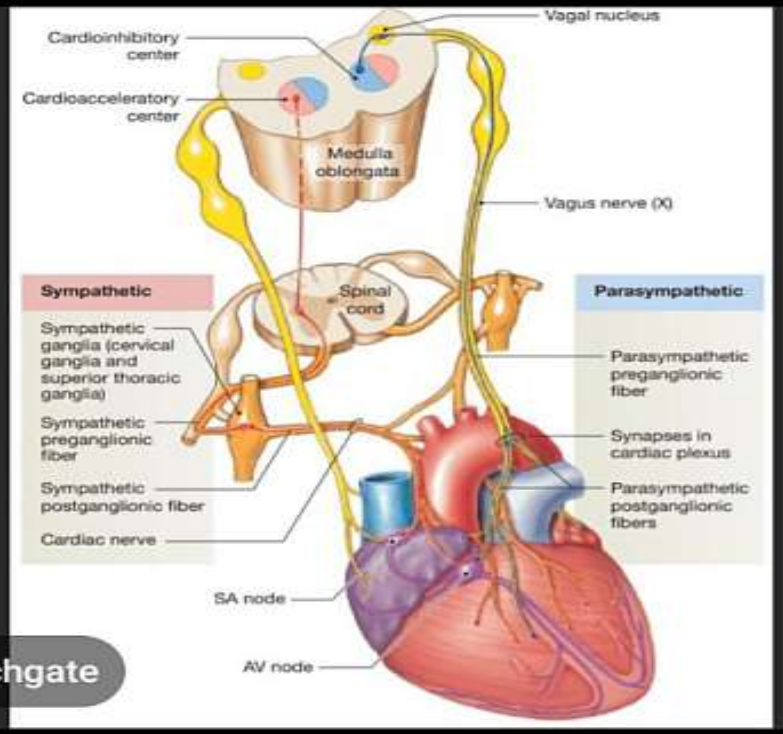
- Preganglionic fibers arise from the spinal cord segments **T1–T5** (intermediolateral cell column).
- They synapse in cervical and upper thoracic sympathetic ganglia.
- Postganglionic fibers reach the heart through cardiac nerves.

Effects on the heart:

- Increases heart rate (positive chronotropic effect).
- Increases force of contraction (positive inotropic effect).
- Increases conduction velocity (positive dromotropic effect).
- Increases excitability.

Neurotransmitter: Noradrenaline (norepinephrine), acting mainly on β1 receptors.

2. Parasympathetic innervation



Origin:

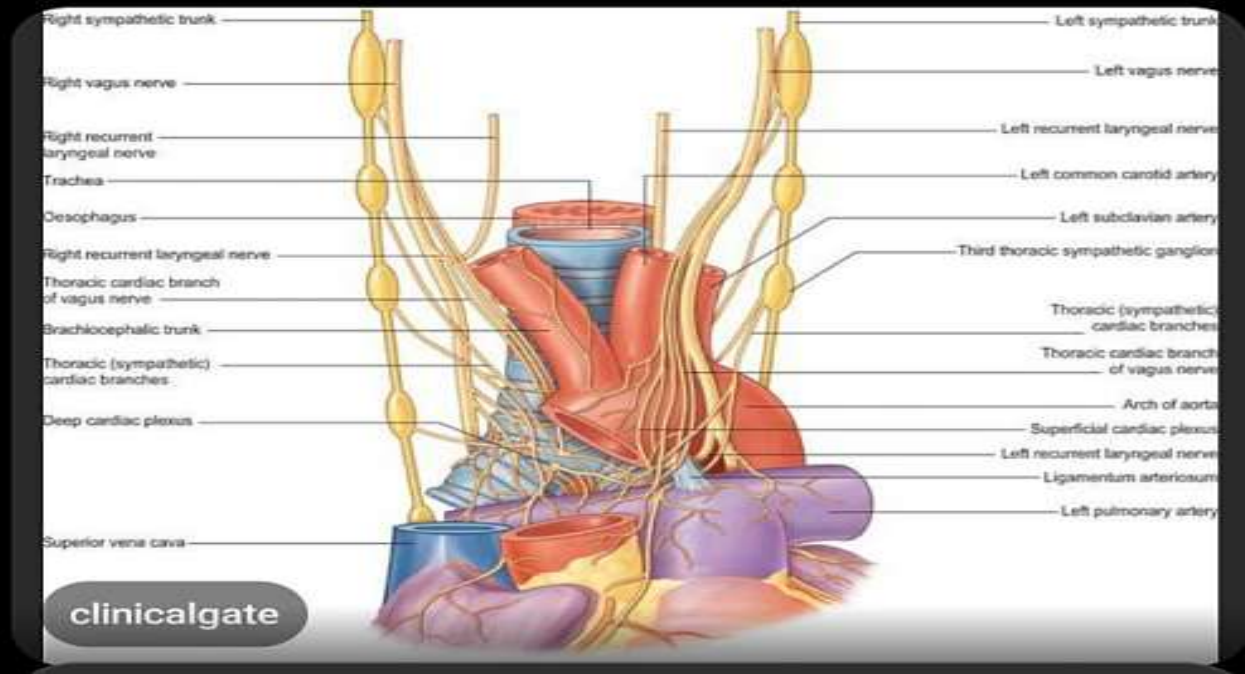
- From the medulla oblongata.
- Fibers travel through the right and left vagus nerves (cranial nerve X).
- Cardiac branches reach the cardiac plexus and synapse in small ganglia near or within the heart.

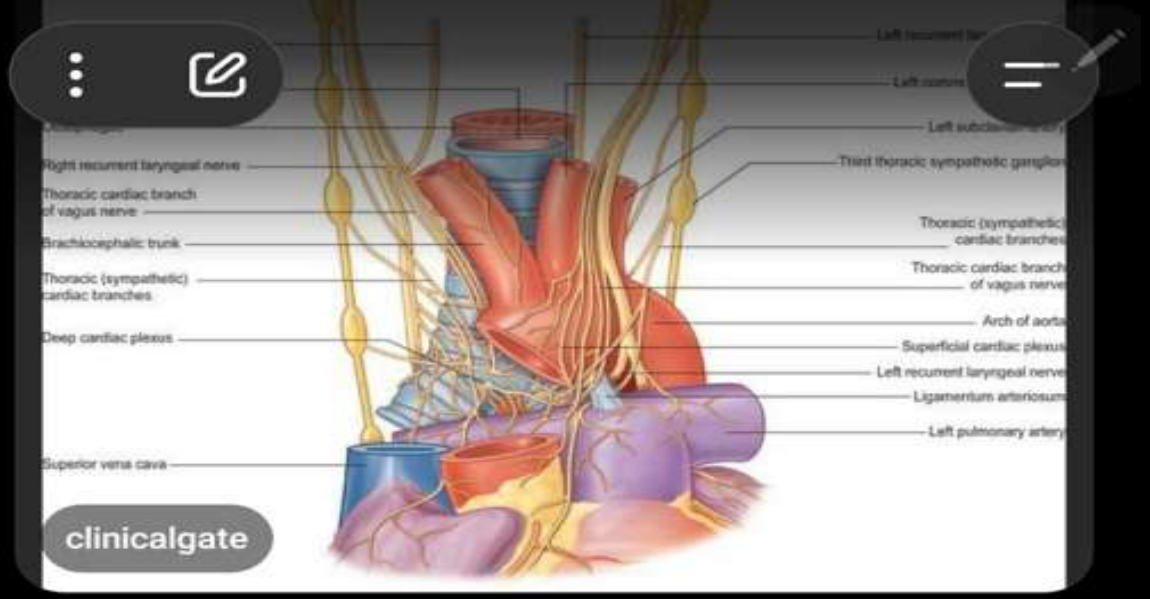
Effects on the heart:

- Decreases heart rate.
- Decreases conduction velocity, especially through the AV node.
- Slightly decreases atrial contraction force.
- Has relatively little direct effect on ventricular contractility.

Neurotransmitter: Acetylcholine, acting on muscarinic M2 receptors.

3. Cardiac plexus





The cardiac plexus is formed by sympathetic and parasympathetic fibers.

It has two parts:

Plexus	Location
Superficial cardiac plexus	Below the aortic arch, anterior to right pulmonary artery
Deep cardiac plexus	Anterior to tracheal bifurcation, posterior to aortic arch

Both plexuses distribute nerves to the heart and coronary vessels.

4. Right versus left vagus nerve

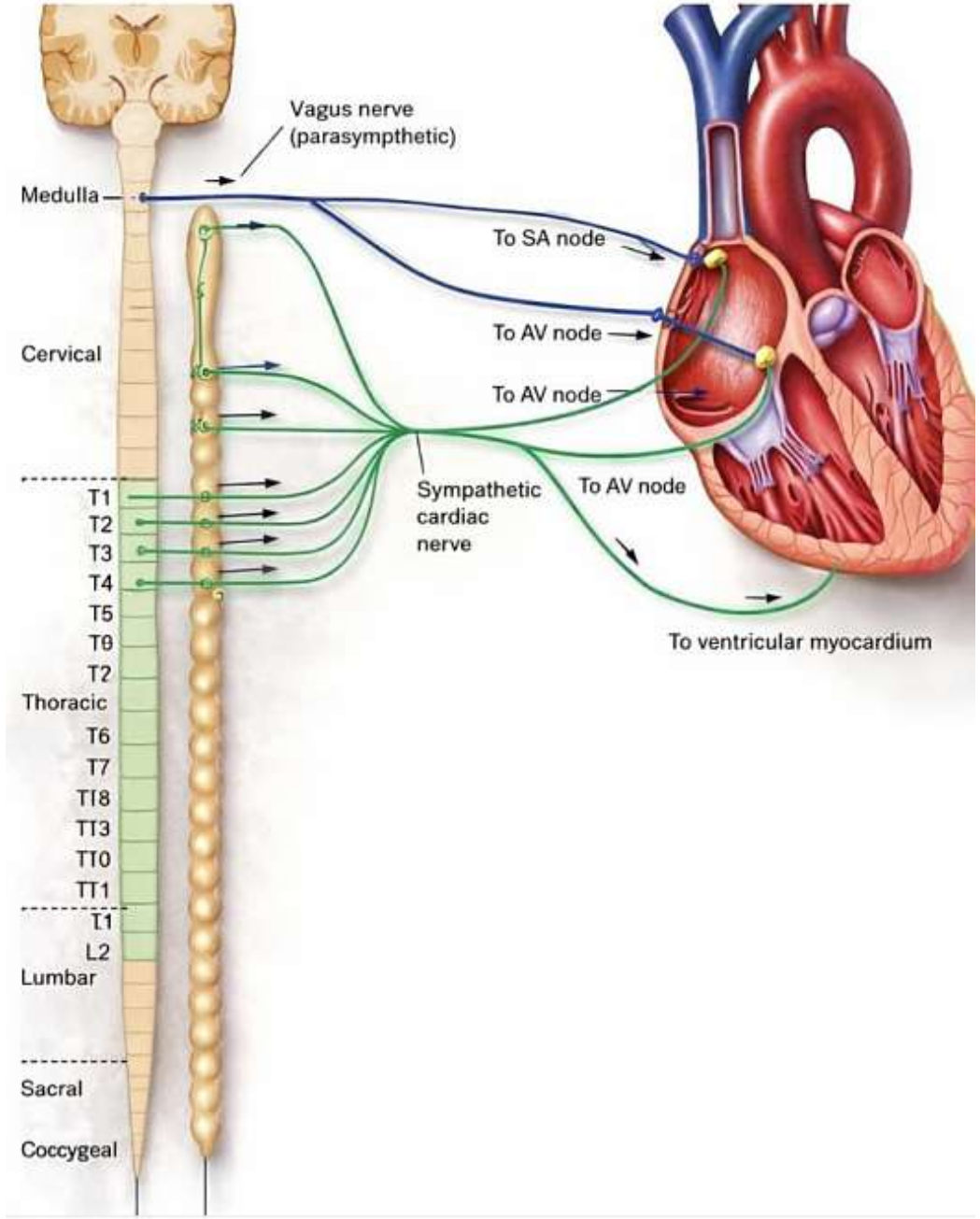
An important surgical and examination point

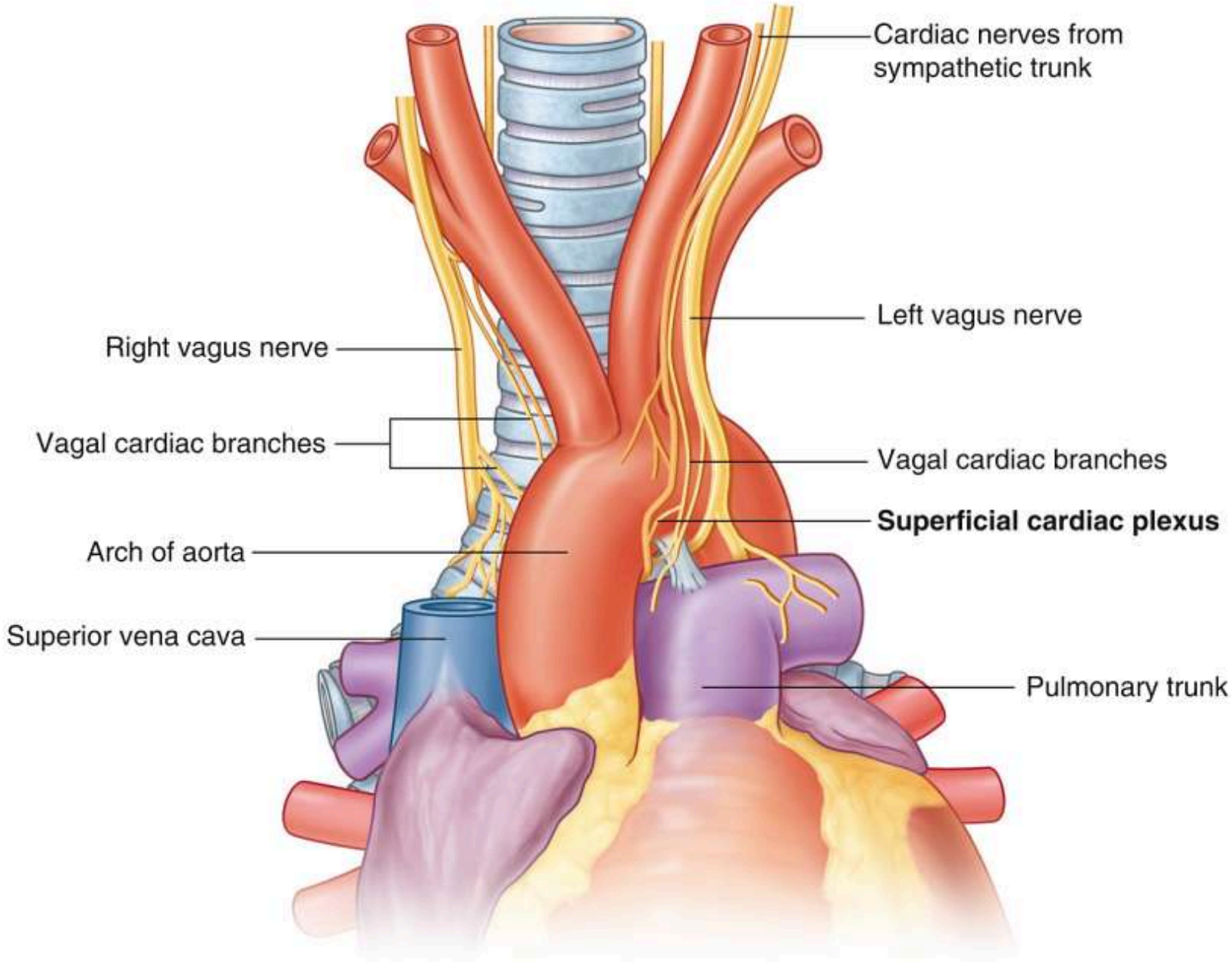
- **Right vagus nerve:** Has a greater influen on the SA node (heart rate).
- **Left vagus nerve:** Has a greater influenc on the AV node (AV conduction).

This is a predominant tendency, not an absolute separation.

The effect of Vagus Nerve on Heart ✅

#Lifestyle




Cardiac nerves from sympathetic trunk
Left vagus nerve
Right vagus nerve
Vagal cardiac branches
Vagal cardiac branches
Superficial cardiac plexus
Arch of aorta
Superior vena cava
Pulmonary trunk
A
Cardiac nerves from sympathetic trunk

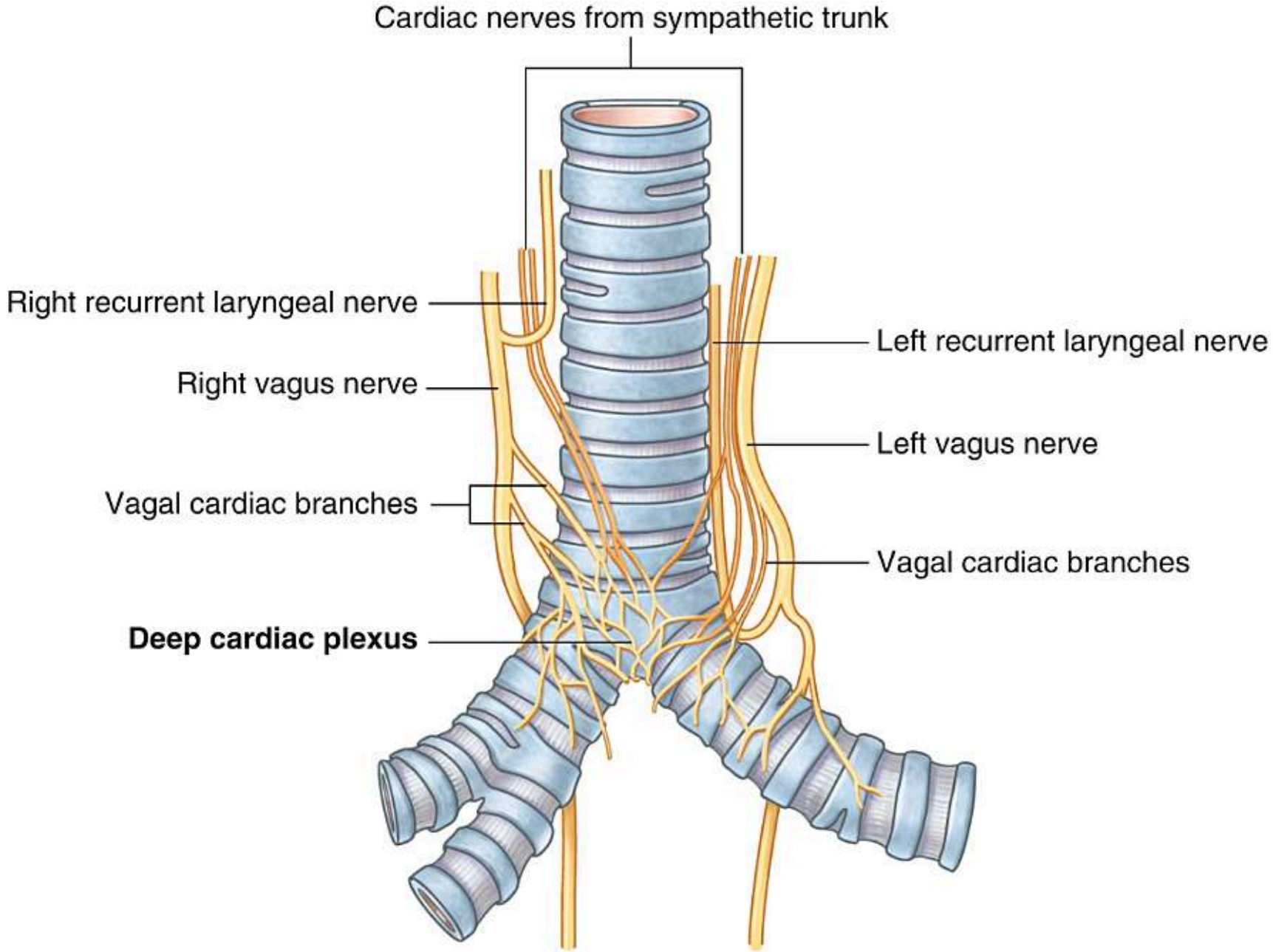


Fig. 3.87 Cardiac Plexus. (**A**) Superficial. (**B**) Deep.

Parasympathetic innervation

Stimulation of the parasympathetic system:

- decreases heart rate,
- reduces force of contraction, and
- constricts the coronary arteries.

The preganglionic parasympathetic fibers reach the heart as cardiac branches from the right and left vagus nerves. They enter the cardiac plexus and synapse in ganglia located either within the plexus or in the walls of the atria.

Sympathetic innervation

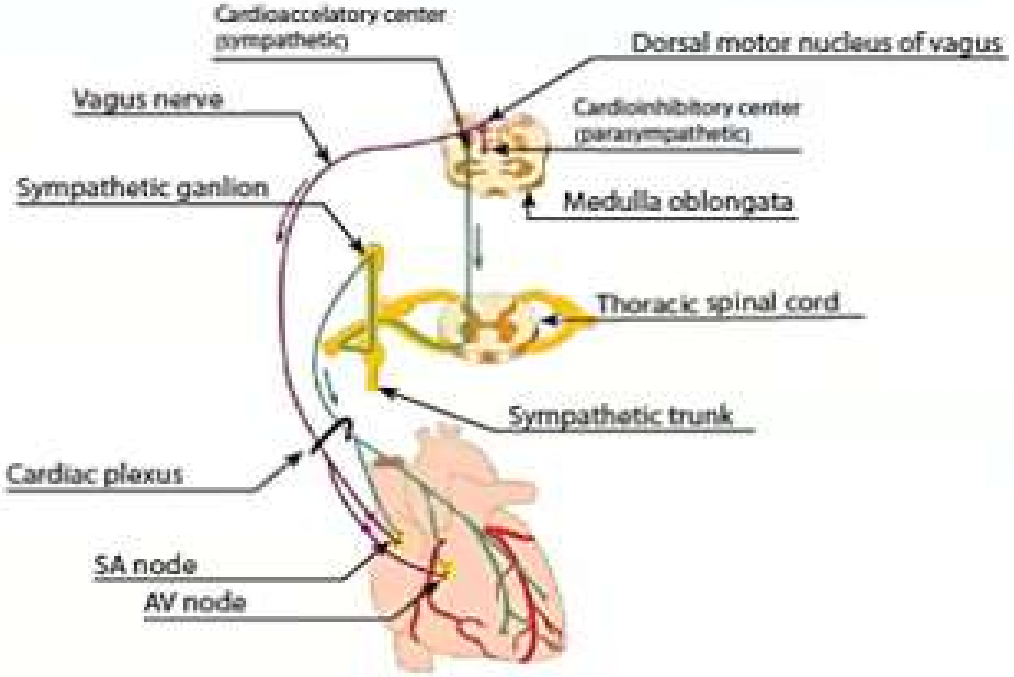
Stimulation of the sympathetic system:

- increases heart rate, and
- increases the force of contraction.

Nerve supply of the heart

Sympathetic innervation

- They are postganglionic fibers.
- They originate from cervical and upper thoracic sympathetic ganglia.
- They reach SAN, AVN, cardiac muscle, and coronary arteries.
- They increase the HR, force of contraction, and cause VD of coronaries.
- They accompany afferent fibers carrying pain sensation from heart.



Parasympathetic innervation

- They are preganglionic fibers.
- They are the cardiac branches of both vagus nerves.
- They reach SAN, AVN, and coronary arteries.
- They decrease the HR, force of contraction, and cause VC of coronaries.
- They accompany afferent fibers for cardiac reflexes.

